DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY UTTAR PRADESH, LUCKNOW



Evaluation Scheme & Syllabus For

B.Tech. 4th Year

MINING ENGINEERING

[Effective from the Session: 2026-27}

7TH semester														
SN	Subject Code	Subject	Type	Periods			End Semester Examination (ESE)						Total SW+ESE	Credit Cr
				L	T	P	CT	TA	CT+ TA	PS	TE	PE		
1	BMN-701	Mine Machinery	T	3	0	0	20	10	30		70		100	3
2	BMN-***	Departmental Elective-4	T	3	0	0	20	10	30		70		100	3
3	BOE***	Open Elective-2	T	3	0	0	20	10	30		70		100	3
4	BMN-751	Mine Machinery Lab	P	0	0	2				50		50	100	1
5	BMN-752	Mini Project or Internship Assessment*	P	0	0	4				100			100	2
6	BMN-753	Project-1	P	0	0	10				150			150	5
7	BMN-754	Startup and Entrepreneurial Activity Assessment#	T	0	0	4				100			100	2
		Total		**9**	**0**	**20**							**750**	**19**

**The Mini Project or internship (5-6 weeks) conducted during summer break after VI semester and will be assessed during VII semester.*

#The Startup and Entrepreneurial Activity Assessment will be done in 7th semester under which a student will have to undergo a startup/entrepreneurship activity of at least 60 hours till 6th semester.

8TH semester														
SN	Subject Code	Subject	Type	Periods			End Semester Examination (ESE)						Total SW+ESE	Credit Cr
				L	T	P	CT	TA	CT+TA	PS	TE	PE		
1	BOE***	Open Elective-3	T	3	0	0	20	10	30		70		100	3
2	BOE***	Open Elective-4	T	3	0	0	20	10	30		70		100	3
3	BMN-851	Project-2	P	0	0	18				100		350	450	10
		Total		**6**	**0**	**18**							**650**	**16**

Note: The Internal Assessment of MOOCs will be done by the respective institute and the external Assessment (End Semester Examination) will be done by the University.

Departmental Electives

7th Semester	**Departmental Elective-4**	**BMN 071**	**BMN 072**	**BMN 073**
		Mine Economics	Mine Safety Engineering	Mine Automation & Data Analytics

7th Semester

Subject Code: BMN701	Mine Machinery	L T P : 3 0 0	Credits: 3

CO	Course Outcome: Student will be able to	Bloom Level
CO1	Explain the principles of generation and transmission of mechanical, hydraulic and pneumatic power in mines; analyse compressors and hydraulic transmission elements.	K2
CO2	Specify wire ropes and design rope haulages, locomotive haulages, belt and chain conveyors and aerial ropeways with productivity and power calculations.	K3
CO3	Describe drum and Koepe winding systems and compute winding parameters (torque–time, power–time and horse-power requirement).	K3
CO4	Select appropriate face and development machinery (drills, cutter loaders, shearers, road headers, LHD/SDL, continuous miners and surface mining equipment) for given geo-mining conditions.	K4
CO5	Select and analyse mine pumps; apply comminution theories and evaluate crushers and grinding mills used in mineral processing.	K3

UNIT I [07 Hours]

Power Generation & Transmission in Mines

Mechanical Power Transmission: belt drives, chain drives and gear drives; couplings (rigid, flexible, fluid); clutches and brakes used in mining machinery; basic selection criteria for mining duty.

Hydraulic Power Systems: fundamentals of hydrostatic and hydrodynamic transmission; hydraulic fluids; hydraulic pumps (gear, vane, piston), motors, cylinders and accumulators; directional, pressure and flow-control valves; hydraulic couplings and torque converters; applications in mines.

Pneumatic Power and Compressed Air: principles of compression; single-stage and multi-stage reciprocating compressors; rotary and screw compressors (overview); intercoolers, after-coolers, receivers and unloading arrangement; layout of compressed-air pipelines; transmission losses; in-bye compressors.

UNIT II [08 Hours]

Wire Ropes & Haulage Systems

Wire Ropes: applications in surface and underground mines; construction, lays, classification (round-strand, locked-coil, non-stranded, flattened-strand); factor of safety; examination, testing, care and maintenance; capping (white-metal and resin) and splicing; rope capels; rope-changing procedure.

Track Haulage: direct, endless, main-and-tail and gravity haulage systems; pull and power calculations; safety devices — Lilly controller, stop blocks, runaway switches, drag, manholes and refuge holes.

Locomotive Haulage: trolley-wire, diesel, battery and flameproof locomotives; track laying and maintenance; mine cars and tubs; productivity calculations for haulage systems.

UNIT III [08 Hours]

Conveyors, Aerial Ropeways & Hoisting Systems

Belt Conveyors: construction and main components — drums, idlers, pulleys, fabric and steel-cord belting; drive arrangements; tensioning devices (gravity, screw, hydraulic take-up); loading and discharge arrangements; safety and sequence-control devices; capacity and power calculations; splicing and maintenance.

Chain Conveyors: armoured face conveyor (AFC) and panzer conveyor — construction, drive units, chain types and applicability at longwall faces; scraper-chain and shaker conveyors — construction and application.

Aerial Ropeways: monocable and bicable systems; construction, layout, rope-tensioning and applications.

Hoisting — Shaft Arrangement: headgear, shaft fittings; cages, skips and their suspension gear; detaching hooks; principal safety devices.

UNIT IV [09 Hours]

Winding Systems and Face / Development Machinery

Drum Winders: cylindrical and conical drums; winding cycles; mechanical and electrical braking (dynamic, regenerative and plugging).

Electric Winders: AC and DC drives; speed control (overview); safety circuits and interlocks.

Koepe Winding: ground-mounted and tower-mounted Koepe winders; single and multi-rope friction winding; comparison with drum winding.

Winding Calculations: torque–time and power–time diagrams; horse-power requirement; selection of winder for given duty cycle.

Drilling Machines: jack-hammers, jumbos, drifters and wagon drills; rotary coal drills; DTH and rotary blast-hole drills for opencast mining; selection criteria and productivity assessment.

Coal Cutting & Cutter Loaders: conventional coal cutters and ploughs; single-ended and double-ended ranging-drum shearer (DERDS); interface with powered supports.

Road Headers: boom-type road headers; rock-tool interaction; selection based on rock strength.

Underground Loading & Transport: rocker shovel, gathering-arm loader, side-discharge loader (SDL), load-haul-dump (LHD), shuttle cars and underground trucks; productivity calculation; continuous miner systems.

Surface Mining Equipment: rope shovel, hydraulic excavator, dragline (with range-diagram concept), bucket-wheel excavator, continuous surface miner, scrapers, dozers, front-end loaders; productivity calculation for shovel–dumper combination.

UNIT V [08 Hours]

Mine Drainage Pumps and Comminution Machinery

Mine Drainage: sources of mine water; drainage planning; dewatering of opencast and underground mines.

Types of Mine Pumps: reciprocating, centrifugal (single-stage and multi-stage), turbine, submersible, sinking pumps and sludge pumps; pump fittings and ranging arrangements.

Pump Characteristics: head–discharge, power and efficiency curves; net positive suction head (NPSH) and cavitation; series and parallel operation; pump selection for given duty.

Comminution Theories: Rittinger's, Kick's and Bond's laws; energy considerations and grindability indices.

Crushers: jaw, gyratory, cone and roll crushers — construction, operation and selection criteria.

Grinding Mills: ball, rod, tube and pebble mills; critical speed; open and closed-circuit grinding (overview).

Books and References

1. Deshmukh, D. J., Elements of Mining Technology — Vol. I & Vol. III, Denett & Co.
2. Walker, S. C., Mining Machinery.
3. Deshmukh, R. T. and Deshmukh, D. J., Winning and Working of Coal.
4. Singh, R. D., Principles and Practices of Modern Coal Mining, New Age International.
5. Hartman, H. L. (Ed.), SME Mining Engineering Handbook, 3rd Edition, SME.
6. Peng, S. S., Longwall Mining, CRC Press.
7. Wills, B. A. and Finch, J., Wills' Mineral Processing Technology, 8th Ed., Butterworth-Heinemann.
8. Cummins & Given, SME Mining Engineering Handbook Vol. I & II.
9. OEM manuals — Atlas Copco, Sandvik, Caterpillar, Komatsu, BEML (latest editions).

DEPARTMENTAL ELECTIVE – 4

Subject Code: MN071	Mine Economics	L T P : 3 0 0	Credits: 3

Course Outcome: Student will be able to		Bloom Taxonomy
CO 1	Classify mineral resources and reserves as per UNFC; carry out sampling and grade-reserve estimation for mineral deposits.	K2
CO 2	Apply standard methods of mine valuation (Hoskold's, Morkill's, present-value) and compute depreciation and capital recovery for mining assets.	K3
CO 3	Evaluate mining investment proposals using discounted and non-discounted cash flow methods and perform sensitivity analysis.	K4
CO 4	Explain principles of financial management of mining enterprises and apply provisions of mineral taxation under MMDR Act and related rules.	K3
CO 5	Prepare profit-and-loss account, balance sheet and cost-of-production statements; demonstrate awareness of National Mineral Policy and sustainable mining.	K3

UNIT I [06 Hours]

Mineral Resources, Reserves and Sampling

Resource and Reserve Classification: concepts; IBM and UNFC frameworks; JORC and CRIRSCO codes; measured, indicated and inferred resources; proved, probable and possible reserves.

Sampling and Statistical Analysis: grab, channel, chip, bulk and drill-core sampling; sample preparation and reduction; salting and precautions; statistical treatment — mean, variance and standard error of grade.

Grade and Reserve Estimation: cut-off grade and break-even grade; reserve estimation by cross-section, polygonal, triangular and inverse-distance methods; introduction to geostatistics and kriging; grade-tonnage curve.

UNIT II [07 Hours]

Mine Valuation

Valuation Fundamentals: objectives and scope of mine valuation; factors affecting mine value; valuation for sale, taxation and mortgage; distinction between mine, accounting and market valuation.

Methods of Mine Valuation: present-value method; Hoskold's, Morkill's and O'Donahue formulae; replacement-cost and comparative methods; relative merits and limitations.

Life and Capacity of a Mine: Taylor's rule for optimum capacity; relationship between reserve, capacity and life; effect of cut-off grade on life and capacity.

Depreciation and Capital Recovery: straight-line, declining-balance, sum-of-years-digits, sinking-fund and production-unit methods; amortization of intangibles; redemption of capital; depletion allowance.

UNIT III [09 Hours]

Discounted Cash Flow Analysis

Time Value of Money: compounding and discounting; present value, future value, annuity and perpetuity; effect of compounding frequency.

Cash Flow Analysis: project cash flow diagrams; project life cycle — exploration, development, production and closure; treatment of escalation, inflation and working capital; tax-affected cash flow.

Investment Appraisal Methods: non-discounted methods — payback period and accounting rate of return; discounted methods — NPV, IRR, discounted payback, profitability index, B/C ratio, modified IRR; conflict between NPV and IRR for ranking.

Sensitivity and Risk Analysis: one-way sensitivity (spider and tornado diagrams); scenario analysis; Monte-Carlo simulation (overview); risk-adjusted discount rate.

Project Selection: ranking of mutually exclusive projects; capital rationing; replacement decisions; case studies on mining-project appraisal.

UNIT IV [09 Hours]

Financial Management and Mineral Taxation

Business Organization and Financing: forms of business organization; memorandum and articles of association; sources of short-term and long-term capital for mining enterprises.

Capital Structure: equity shares, preference shares, debentures, retained earnings and term loans; cost of capital; weighted average cost of capital (WACC); leverage.

Break-Even and Inventory: break-even chart, break-even point, margin of safety and contribution margin; ABC analysis, Economic Order Quantity (EOQ), reorder level and safety stock.

Mineral Taxation: royalty (tonnage and ad-valorem); dead rent; cess — District Mineral Foundation (DMF) and National Mineral Exploration Trust (NMET); income tax on mining; GST impact; MMDR Act 1957 (as amended in 2015 and 2021); Mineral Concession Rules 2016.

Mineral Pricing: factors governing mineral prices; primary versus secondary markets; price volatility and cyclical nature of the mineral economy; long-term contracts versus spot markets; commodity exchanges.

UNIT V [05 Hours]

Financial Statements and Mineral Economics in Practice

Book-Keeping Essentials: double-entry system; journal, ledger and subsidiary books; trial balance; rectification of errors.

Financial Statements: preparation of profit-and-loss account and balance sheet; key financial ratios — current ratio, debt-equity, ROCE and earnings per share.

Cost of Production: cost classification — fixed, variable, direct, indirect and overhead; cost per tonne; cost centres in a mine.

Mineral Economics in Industry: National Mineral Policy 2019; pre-feasibility, feasibility and bankable feasibility studies; sustainable mining and ESG reporting (overview).

Books and References:

- Deshmukh, R. T., Mineral Economics, Lovely Prakashan.
- Chatterjee, K. K., An Introduction to Mineral Economics, Wiley Eastern.
- Hartman, H. L. (Ed.), SME Mining Engineering Handbook, Vol. I (Economic chapters), 3rd Ed., SME.
- Hustrulid, W., Kuchta, M. and Martin, R., Open Pit Mine Planning and Design, Vol. 1: Fundamentals, 3rd Ed., CRC Press.
- Pandey, I. M., Financial Management, Vikas Publishing House.
- Indian Bureau of Mines — Indian Minerals Year Book and National Mineral Inventory (latest issues).
- MMDR Act 1957 (as amended in 2015 and 2021) and Mineral Concession Rules 2016 — Ministry of Mines, Government of India.
- Annual Reports of major Indian mining companies — NMDC, Coal India, Hindustan Zinc, Vedanta (latest available).

Subject Code: BMN-072	Mine Safety Engineering	L T P : 3 0 0	Credits: 3

CO	Course Outcome: Student will be able to	Bloom Level
CO1	Explain the types, causes and investigation of mine accidents, accident statistics, and the essential elements of a mine safety-management system.	K2
CO2	Analyse the hazards arising from mine gases, fires and explosions, and select appropriate detection, monitoring and control measures.	K4
CO3	Apply dust-control, illumination, inundation-control and occupational-health measures to safeguard the health and safety of mine workers.	K3
CO4	Apply fault-tree analysis, probabilistic risk analysis and reliability concepts to identify, assess and manage risks in mines.	K4
CO5	Explain the statutory framework, rescue and recovery practice, and disaster-management measures applicable to Indian mines.	K2

UNIT I [08 Hours]

Mine Accidents, Accident Analysis and Safety Management

Mine Accidents: occupational hazards of mining; accidents in mines — different types and classification; accident investigation; accident proneness; basic causes of accident occurrence; place-wise and cause-wise analysis.

Accident Statistics & Costs: frequency and severity rates of accidents; accident-data analysis and assessment; relation between accidents and efficiency; direct and indirect cost of accidents.

Safety Management: creating and maintaining safety awareness; ZAP and MAP; job safety analysis; safety meetings and committees; management's role and responsibility; introduction to risk-based safety and health management systems; safety management plan; measures for improving safety in mines.

UNIT II [08 Hours]

Underground Hazards I — Mine Gases, Fires and Explosions

Mine Gases: composition of the mine atmosphere; production, properties and physiological effects of mine gases; sampling and analysis of mine air; methane content and methane layering; methane drainage; flame safety lamp and methanometers; radon and its daughter products; continuous monitoring of gases.

Mine Fires: causes and classification of mine fires; spontaneous combustion — mechanism, stages and susceptibility indices; factors affecting spontaneous heating; detection and prevention; dealing with mine fires by direct and indirect methods; fire stoppings and sealing-off; re-opening of sealed-off areas; fires in coal stacks and waste dumps.

Mine Explosions: firedamp and coal-dust explosions — causes, mechanism and prevention; explosive limits; stone-dust and water barriers; explosions in opencast workings; investigation after an explosion.

UNIT III [08 Hours]

Underground Hazards II — Dust, Inundation, Illumination and Occupational Health

Airborne Dust: generation, dispersion, measurement and control of airborne respirable dust; sampling instruments and methods, duration and interval of sampling; prevention and suppression of dust; physiological effects of dust and dust-related diseases (pneumoconiosis, silicosis).

Inundation: causes and prevention of inundation; precautions and techniques in approaching old workings; dewatering of waterlogged workings; safety boring apparatus and pattern of holes; design and construction of water dams.

Illumination & Occupational Health: standards of illumination for underground and opencast workings; cap lamps; lamp-room layout and organisation; photometry and illumination survey; comfort conditions, first aid and ambulance; miners' occupational diseases — causes, nature and preventive measures; preventive medical examinations and compensable diseases.

UNIT IV [08 Hours]

Risk Assessment and Safety-Engineering Techniques

Risk Assessment: principles of risk assessment; hazard identification; risk and hazard control; risk and hazard evaluation; data collection for identified health risks; exposure assessment and risk characterisation; methods of risk assessment.

Fault-Tree Analysis: introduction and methodology; symbols and Boolean techniques; qualitative analysis; computerised methods; statistical analysis and safety information; systems design.

Reliability & Safety Audit: probabilistic risk analysis; reliability evaluation; human factors; safety culture; safety audit and its application to mine-safety problems.

UNIT V [08 Hours]

Mine Legislation, Rescue, Recovery and Disaster Management

Mine Legislation: the Mines Act, 1952; the Coal Mines Regulations, 2017; the Metalliferous Mines Regulations, 1961; statutory controls and directions; spot and regular inspections; enforcement of standards and penalties for violations; role of the Directorate General of Mines Safety (DGMS); safety campaigns and Safety Week.

Rescue & Recovery: rescue equipment and its uses; classification of rescue apparatus; resuscitation; rescue stations and rescue rooms; organisation of rescue work; emergency preparedness and response systems; Mines Rescue Rules and Mines Vocational Training Rules.

Disaster Management: identification of causes of mine disasters; preventive action; disaster management and mitigation; typical cases of mine disasters in India; recommendations of inquiry committees on safety and health issues.

Books and References

1. Deshmukh, D. J., *Elements of Mining Technology, Vol. 2*, Denett & Co., Nagpur.
2. Misra, G. B., *Mine Environment and Ventilation*, Oxford University Press.
3. DeReamer, R., *Modern Safety Practices*, John Wiley and Sons.
4. McPherson, M. J., *Subsurface Ventilation and Environmental Engineering*, Chapman & Hall.
5. Prasad, S. D. and Rakesh, *Legislation in Indian Mines — A Critical Appraisal*, Vols. I & II.
6. Kaku, L. C., *CMR-2017 & MMR-1961*; *Mines Act-1952 & Mines Rules-1955*.

7. Kejriwal, S. J., *Mine Accidents*.
8. Van Zyl, D. A., Koval, M. and Li, T. M. (Eds.), *Risk Assessment / Management Issues in the Environmental Planning of Mines*, SME, 1992.

Subject Code: BMN-073	Mine Automation & Data Analytics	L T P : 3 0 0	Credits: 3

CO	Course Outcome: Student will be able to	Bloom Level
CO1	Explain the essential elements, principles and strategies of mine automation and the working of autonomous mining, fleet-management and remote-operation systems.	K2
CO2	Make use of automated communication, tracking, sensing and virtual-reality technologies for implementing automation and ensuring safe operations in mines.	K3
CO3	Apply descriptive and inferential statistics and fundamental machine-learning concepts to analyse mining data.	K3
CO4	Analyse mining data using supervised learning, artificial neural networks and deep-learning techniques.	K4
CO5	Apply unsupervised learning, big-data analytics and artificial intelligence to mining decision-making and optimisation problems.	K4

UNIT I [08 Hours]

Foundations of Mine Automation and Autonomous Systems

Automation Fundamentals: essential elements of an automated system; automation in the mine production system; principles and strategies of automation; introduction to automation productivity and levels of automation.

Autonomous Mining Systems: autonomous haulage systems (AHS); automated drilling systems; tele-remote and autonomous operating modes; selection considerations for autonomous equipment.

Fleet Management & Remote Operations: Fleet Management Systems — Truck Dispatch System (TDS), Computerised Maintenance Management System (CMMS) and Enterprise Resource Planning (ERP) for the mining industry; mine remote operations and control; mining robotics and “armchair mining”; remote operating centres.

UNIT II [08 Hours]

Sensing, Communication, Tracking and Virtual-Reality Technologies

Automated Communication & Tracking Technologies: proximity-detection systems; GNSS/UPS positioning; radar systems; RFID and geo-fencing; CCD cameras and machine vision; image processing; SCADA architecture and its applications in mines.

Virtual Reality & Simulation: virtual-reality applications in mining; mining-equipment concept development; mine-safety training applications; mining-operation simulations and visualisation of the mining process for safe operations.

UNIT III [08 Hours]

Statistical Foundations and Introduction to Machine Learning

Descriptive Statistics: measures of central tendency and dispersion; data summarisation and visualisation; probability and probability distributions (discrete and continuous) relevant to mining data.

Inferential Statistics: sampling and estimation; hypothesis testing; regression analysis; analysis of variance (ANOVA); interpretation of results for mining problems.

Introduction to Machine Learning: basic concepts, terminology and the learning workflow; supervised, unsupervised and reinforcement learning; the data-analytics pipeline for mining applications.

UNIT IV [08 Hours]

Supervised Learning, Neural Networks and Deep Learning

Supervised Learning: classification and regression; feature engineering; training, validation and testing; the bias–variance trade-off; model-evaluation metrics.

Artificial Neural Networks: the perceptron and multilayer networks; activation functions; the back-propagation algorithm; network training and regularisation.

Deep Learning: deep network architectures (overview); representation and feature learning; representative applications of deep learning to mining and geoscience data.

UNIT V [08 Hours]

Unsupervised Learning, Big-Data Analytics and AI Applications in Mining

Unsupervised Learning: clustering techniques; pattern and structure discovery in unlabelled mining data; dimensionality reduction (overview).

Big-Data Analytics: characteristics and challenges of big data; data acquisition, storage and management; analytics frameworks and pipelines for large-scale mining data.

Applications in Mining: application of big-data analytics and artificial intelligence in mining — predictive maintenance, production and energy optimisation, safety analytics and decision-support systems.

Books and References

1. Hastie, T., Tibshirani, R. and Friedman, J., The Elements of Statistical Learning, Vol. 2, Springer, New York, 2009.
2. Montgomery, D. C. and Runger, G. C., Applied Statistics and Probability for Engineers, John Wiley & Sons, 2010.
3. Almgren, G., Kumar, U. and Vagenas, N., Mine Mechanization & Automation, 1st Edition.
4. O'Shea, J. and Polis, M., Automation in Mining, Mineral and Metal Processing, Proceedings of the 3rd IFAC Symposium, Montreal, Canada, 18–20 August 1980.
5. Moreira, J., Carvalho, A. and Horvath, T., A General Introduction to Data Analytics, Wiley, 2019.
6. Runkler, T. A., Data Analytics: Models and Algorithms for Intelligent Data Analysis, Vieweg+Teubner Verlag, 2012.

LAB

Subject Code: BMN751	Mine Machinery Lab	L T P : 0 0 2	Credits: 1

CO	Course Outcome: Student will be able to
CO1	Identify and examine wire ropes, rope capels and rope-end attachments used in mines and explain their maintenance procedure.
CO2	Demonstrate the construction and operation of haulages, conveyors and hoisting installations through working models.
CO3	Explain the constructional features and operating principles of face and development machinery (SDL, LHD, AFC, shearer).
CO4	Analyse the working of mine pumps, compressors and comminution equipment and select suitable machinery for given mining conditions.

List of Experiments (any TEN to be performed)

1. Study of different types of rope capels (white-metal, wedge and resin) and rope-end attachments.
2. Study of wire-rope splicing techniques and rope-changing procedure in haulage and winding installations.
3. Study of safety devices on rope haulages — Lilly controller, stop block, runaway switch, drag and manhole.
4. Study of cage suspension gear and detaching hook (King–Hepburn type).
5. Study of Koepe winding arrangement and its comparison with drum winding (cut-section model).
6. Study of constructional features and tensioning arrangement of a belt conveyor.
7. Study of monocable and bicable aerial ropeway systems through working models.
8. Study of armoured face conveyor (AFC) used at longwall faces.
9. Study of side-discharge loader (SDL) and load-haul-dump (LHD) machine.
10. Study of double-ended ranging-drum shearer (DERDS) and its interface with powered supports.
11. Study of constructional features and working of single-stage and multi-stage reciprocating compressors.
12. Study of hydraulic coupling and torque converter (cut-section model).
13. Study of mine pumps — turbine, multi-stage centrifugal and submersible pumps; study of balancing disc.
14. Study of construction and operation of jaw crusher and roll crusher (laboratory model).
15. Study of ball mill / tube mill operation; determination of critical speed (calculation exercise).
16. Selection case-study: choose appropriate haulage / hoisting / shovel–dumper combination for a given mining scenario.

Note:

• At least 50% of experiments should be performed on actual / cut-section / working models; the remainder may use demonstrations, simulation software or OEM videos.

• Industrial visit to at least one underground mine and one opencast mine is recommended.

• Practical Sessional (PS) — 50 marks: attendance, lab record, viva and continuous assessment. Practical End-Semester Exam (PE) — 50 marks: experiment performance, sketching and viva-voce.